

MSC2050 Discrete Mathematics, Week 11

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Relations and their properties

(Paragraph 9.1)

Let A and B be sets. A **binary relation** from A to B is a subset R of $A \times B$. We use the notation $a R b$ to denote that $(a, b) \in R$ and $a \not R b$ to denote that $(a, b) \notin R$.

Moreover, when (a, b) belongs to R , a is said to be related to b by R .

A is the set of students in the school.

B is the set of courses.

Let's R consists of all pairs (a, b) , where student a is enrolled in the course b .

Assume that Goodfriend and Sherman are enrolled in the course CS518.

Then we write $(\text{Goodfriend}, \text{CS518}) \in R$ and $(\text{Sherman}, \text{CS518}) \in R$

or Goodfriend R CS518 and Sherman R CS518.

$(\text{Sherman}, \text{CS510}) \notin R$ means that Sherman does not enrolled in the course CS510.

Similarly for Sherman $\not R$ CS510.

Let $R = "$ $\leq$ " the relation from $\mathbb{Z}$ to $\mathbb{Z}$.

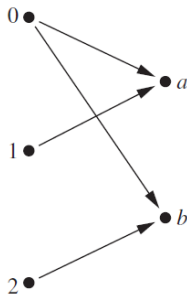
Then $1 \leq 4$ and $5 \not\leq -3$.

Let $A = \{0, 1, 2\}$, $B = \{a, b\}$ and $R = \{(0, a), (0, b), (1, a), (2, b)\}$

Then $0 R a$ and $0 R b$ for instance, but $1 \not R b$.

Displaying relations

If $A = \{0, 1, 2\}$, $B = \{a, b\}$ and $R = \{(0, a), (0, b), (1, a), (2, b)\}$ then R can be displayed using arrows or in table:



R	a	b
0	×	×
1	×	
2		×

Any function gives a relation (graph of this function), but not any relation gives a function.

Relations on a set

A **relation on a set** A is a relation from A to A .

Let $A = \{1, 2, 3, 4\}$ and $R = \{(a, b) \mid a \text{ divides } b\}$ be the relation on A .
Then $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$.

Consider relations on the set of integers:

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R_4 = \{(a, b) \mid a = b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

Then $(1, 1)$ belongs to R_1, R_3, R_4, R_6

$(1, 2)$ belongs to R_1, R_6

$(2, 1)$ belongs to R_2, R_5, R_6

$(1, -1)$ belongs to R_2, R_3, R_6

If A is finite with n elements, then the number of all possible relations on A is 2^{n^2} .

Properties of relations

A relation R on a set A is called **reflexive** if $(a, a) \in R$ for every $a \in A$.

Which of the following relations on integers is reflexive?

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R_4 = \{(a, b) \mid a = b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

$$R_7 = \{(a, b) \mid a \text{ divides } b\}$$

Answer: R_1, R_3, R_4, R_7 .

A relation R on a set A is called **symmetric** if $(b, a) \in R$ whenever $(a, b) \in R$, for all $a, b \in A$.

A relation R on a set A such that for all $a, b \in A$, if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$ is called **antisymmetric**.

Which of the following relations on integers is symmetric or antisymmetric?

$R_1 = \{(a, b) \mid a \leq b\}$ is not symmetric but is antisymmetric.

$R_2 = \{(a, b) \mid a > b\}$ is not symmetric but is antisymmetric.

$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$ is symmetric and antisymmetric

$R_4 = \{(a, b) \mid a = b\}$ is symmetric and antisymmetric

$R_5 = \{(a, b) \mid a = b + 1\}$ is not symmetric and not antisymmetric

$R_6 = \{(a, b) \mid a + b \leq 3\}$ is symmetric but not antisymmetric

$R_7 = \{(a, b) \mid a \text{ divides } b\}$ is not symmetric but antisymmetric

A relation R on a set A is called **transitive** if $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$ for all $a, b, c \in A$.

Which of the following relations on integers is transitive?

$R_1 = \{(a, b) \mid a \leq b\}$ is transitive.

$R_2 = \{(a, b) \mid a > b\}$ is transitive.

$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$ is transitive

$R_4 = \{(a, b) \mid a = b\}$ is transitive

$R_5 = \{(a, b) \mid a = b + 1\}$ is not transitive.

$R_6 = \{(a, b) \mid a + b \leq 3\}$ is not transitive ($a = 1, b = 2, c = 1$)

$R_7 = \{(a, b) \mid a \text{ divides } b\}$ is transitive.

Answer Q1

Homework:

- Solve all odd exercises at p. 581 until #43.

Representing relations (matrices)

Let $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3\}$, let also
 $R = \{(1, 2), (1, 3), (2, 2), (3, 1), (3, 2), (4, 2), (4, 3)\}$, then

$$M_R = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

From the matrix representation we can see if the relation is reflexive, symmetric or antisymmetric

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

not relf.
not sym.
not antisym.
not trans.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

relf.
not sym.
antisym.
not trans.

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

not relf.
sym.
not antisym.
not trans.

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

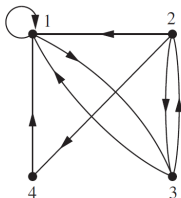
not relf.
not sym.
antisym.
not trans.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

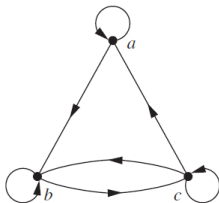
relf.
sym.
antisym.
not trans.

Representing relations (digraphs)

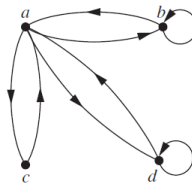
The directed graph of $R = \{(1, 1), (1, 3), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (4, 1)\}$ is



From the digraphs we see if it is reflexive, symmetric, antisymmetric or transitive.



refl. not sym.
not antisym. not trans.



not refl. sym.
not antisym. not trans.

Closure of relations

Reflexive closure of R is the smallest reflexive relation which contains R

Reflexive closure of $R = \{(1, 1), (2, 3), (2, 4), (3, 1)\}$ is
 $\{(1, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 3), (4, 4)\}$

Symmetric closure of R is the smallest symmetric relation which contains R

Symmetric closure of $R = \{(1, 1), (2, 3), (2, 4), (3, 1)\}$ is
 $\{(1, 1), (1, 3), (2, 3), (2, 4), (3, 1), (3, 2), (4, 2)\}$

Transitive closure of R is the smallest transitive relation which contains R

Transitive closure of $R = \{(1, 1), (2, 3), (2, 4), (3, 1)\}$ is
 $\{(1, 1), (2, 1), (2, 3), (2, 4), (3, 1)\}$

Answer Q2-Q4

Warshall's algorithm

procedure Warshall ($M_R : n \times n$ zero-one matrix)

$W := M_R$

for $k := 1$ **to** n

for $i := 1$ **to** n

for $j := 1$ **to** n

$w_{ij} := w_{ij} \vee (w_{ik} \wedge w_{kj})$

return W

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & \color{blue}{1} \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ \color{blue}{1} & 0 & 1 & \color{blue}{1} \end{bmatrix} \rightarrow \begin{bmatrix} \color{blue}{1} & 0 & \color{blue}{1} & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & \color{blue}{1} & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Example

A computer network has data centers in Boston, Chicago, Denver, Detroit, New York, and San Diego. There are direct, one-way telephone lines from Boston to Chicago, from Boston to Detroit, from Chicago to Detroit, from Detroit to Denver, and from New York to San Diego. Let R be the relation containing (a, b) if there is a telephone line from the data center in a to that in b . How can we determine if there is some (possibly indirect) link composed of one or more telephone lines from one center to another?

Homework:

- Solve exercises # 1,3,19,21,23,25,27 at p. 596-597.
- Solve exercises # 1,3,5,7,9,11,25,27,29 at p. 606-607.

Equivalence relations

(Paragraph 9.5)

A relation on a set A is called an **equivalence relation** if it is reflexive, symmetric, and transitive.

Two elements a and b that are related by an equivalence relation are called **equivalent**.

The notation $a \sim b$ is used to denote that a and b are equivalent elements with respect to a particular equivalence relation.

If $R = \{(a, b) \mid a = b\}$ is equivalence relation and $a \sim b$ if and only if $a = b$.

If $R = \{(a, b) \mid a = b \text{ or } a = -b\}$ is equivalence relation and $a \sim b$ if and only if $a = b$ or $a = -b$.

Let aRb be related if $a - b$ is an integer. Then R is equivalence relation (proof).

Let $R = \{(a, b) \mid a \equiv b \pmod{m}\}$, then R is equivalence relation (proof).

Two identifiers (names of variables in some programming language) are related by relation R_8 if when they agree with the first 8 characters. Then R_8 is equivalence relation (proof).

R_3 is equivalence relation on the set of bit strings such that aR_3b if a and b start with the same three bits (proof).

Why $R = \{(a, b) \mid a \text{ divides } b\}$ is not equivalence relation?

Answer Q5

Equivalence Classes and Partitions

Let R be an equivalence relation on a set A . The set of all elements that are related to an element a of A is called the **equivalence class** of a .

The equivalence class of a with respect to R is denoted by $[a]_R$. So

$$[a]_R = \{s \mid (a, s) \in R\}.$$

If $R = \{(a, b) \mid a = b\}$, then $[a]_R = \{a\}$.

Let aRb be related if $a - b$ is an integer. Then

$$[5.2]_R = \{\dots, -1.8, -0.8, 0.2, 1.2, 2.2, \dots\}$$

Let $R = \{(a, b) \mid a \equiv b \pmod{7}\}$, then

$$[5]_R = \{\dots, -9, -2, 5, 12, 19, \dots\}.$$

$$[\text{number_of_something}]_{R_8} = \{\text{number_of_something}, \text{number_of_something} + 8, \text{number_of_something} + 16, \dots\}$$

$$[110]_{R_3} =$$

$$\{110, 1100, 1101, 11000, 11010, 11001, 11011, 1100000, \dots\}.$$

Answer Q6

Theorem

Let R be an equivalence relation on a set A . These statements for elements a and b of A are equivalent:

- (i) aRb
- (ii) $[a] = [b]$
- (iii) $[a] \cap [b] \neq \emptyset$

proof:

(i) $\rightarrow$ (ii). Assume that aRb . We will prove that $[a] = [b]$ by showing $[a] \subset [b]$ and $[b] \subset [a]$. Suppose $c \in [a]$. Then aRc . Because aRb and R is symmetric, we know that bRa . Furthermore, because R is transitive and bRa and aRc , it follows that bRc . Hence, $c \in [b]$. This shows that $[a] \subset [b]$. The proof that $[b] \subset [a]$ is similar.

(ii) $\rightarrow$ (iii). Assume that $[a] = [b]$. It follows that $[a] \cap [b] \neq \emptyset$.

(iii) $\rightarrow$ (i). Suppose that $[a] \cap [b] \neq \emptyset$. Then there is an element c with $c \in [a]$ and $c \in [b]$. In other words, aRc and bRc . By the symmetric property, cRb . Then by transitivity, because aRc and cRb , we have aRb .

We have that $\bigcup_{a \in A} [a]_R = A$ and if $a \not\sim b$ then $[a] \cap [b] = \emptyset$.

Partition of a set

We have that $\bigcup_{a \in A} [a]_R = A$ and if $a \not\sim b$ then $[a] \cap [b] = \emptyset$.

Theorem

Let R be an equivalence relation on a set A .

Then the equivalence classes of R form a **partition** of A .

What is the partition of

$R = \{(a, b) \mid a \equiv b \pmod{4}\}$ on the set of integers.

R_3 on the set of bit strings.

Theorem

Conversely, given a partition $\{A_i \mid i \in I\}$ of the set A , there is an equivalence relation R that has the sets A_i , $i \in I$, as its equivalence classes.

List the ordered pairs in the equivalence relation R produced by the partition $A_1 = \{1, 2, 3\}$, $A_2 = \{4, 5\}$, and $A_3 = \{6\}$ of $A = \{1, 2, 3, 4, 5, 6\}$.

Answer Q7

Homework:

- Solve exercises # 1,3,5,7,9,13,15,19,27,29,37,39,41,43,45,47,57 at p. 615-617.

Partial ordering

(Paragraph 9.6)

A relation R on a set S is called a **partial order** if it is reflexive, antisymmetric, and transitive.

A set S together with a partial order R is called a **partially ordered set**, or **poset**, and is denoted by (S, R) .

The "greater than or equal" relation ($\geq$) is a partial ordering on the set of integers. So $(\mathbb{Z}, \geq)$ is a poset.

$(\mathbb{Z}^+, |)$ is a poset.

$(\mathcal{P}(S), \subseteq)$ is a poset.

The elements $a, b \in S$ are called **comparable** if either aRb or bRa . When a and b are such that neither aRb nor bRa , a and b are **incomparable**.

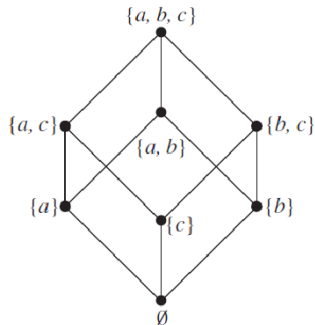
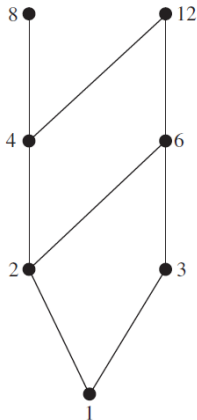
If for all $a, b \in S$, a and b are comparable, then R is called **total order**.

In the poset $(\mathbb{Z}^+, |)$, are the integers 3 and 9 comparable? yes

Are 5 and 7 comparable? no

Which of the relations above is a total order? $\geq$

Hasse Diagrams



$$(\{1, 2, 3, 4\}, \leq), \quad (\{1, 2, 3, 4, 6, 8, 12\}, |), \quad (\mathcal{P}(\{a, b, c\}), \subseteq)$$

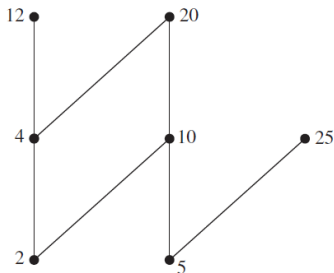
Maximal and Minimal Elements

Let $(S, \preceq)$ be a poset.

The element $a \in S$ is **maximal** if for any $b \in S$ comparable with a , we have $b \preceq a$.

The element $a \in S$ is **minimal** if for any $b \in S$ comparable with a , we have $a \preceq b$.

Which elements of the poset $(\{2, 4, 5, 10, 12, 20, 25\}, |)$ are maximal, and which are minimal?



From the Hasse diagram
the maximal elements are 12, 20, 25
the minimal elements are 2 and 5.

we see that

Topological sorting

procedure topological sort $((S, \preceq)$: finite poset)

$k := 1$

while $S \neq \emptyset$

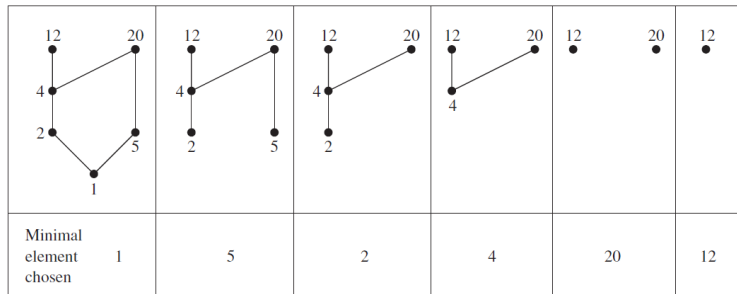
$a_k := a$ minimal element of S

$S := S \setminus \{a_k\}$

$k := k + 1$

return $a_1, a_2, \dots, a_n$

A topological sort of $(\{1, 2, 4, 5, 12, 20\}, |)$ is



Scheduling of projects

A development project requires the completion of seven tasks A, B, C, D, E, F, G . Some of these tasks can be started only after other tasks are finished.

Set $X \preceq Y$ if task Y cannot be started until task X has been completed.

This relation is given as follows $A, C \preceq B$, $B, E \preceq F$, $B \preceq D$, and $D, F \preceq G$. Find an order in which these tasks can be carried out to complete the project.

A topological sort is

Minimal element chosen A	C	B	E	F	D	G

A, C, B, E, F, D, G is one possible order.

Answer Q8

Homework:

- Solve #1,3,5,7,9,11,15,21,23,25,27,33 a) and b), 35 a) and b), 67 at p. 630-633.